

Residual-wrench / parameter diagnostic

Linearized model: $w_{\text{bar}}(\theta + \delta) \approx w_{\text{bar}}(\theta) + J_{\text{bar}} \delta_{\text{bar}}$
Metric: same reference force/torque scaling as deterministic SG second-moment objective
Sample count: 711
Objective Jacobian rank: 13
Absorbable residual second-moment fraction: 1
Irreducible residual second-moment fraction: 0
Per-component absorbable fractions: [1.0, 1.0, 1.0, 1.0, 1.0, 1.0]
Remaining raw wrench mean [Fx,Fy,Fz,Mx,My,Mz]: [-7.68247674e-16 1.42352641e-15 -1.10974192e-14
-1.89562272e-15
1.72321820e-15 1.62410012e-16]
Remaining raw wrench RMS [Fx,Fy,Fz,Mx,My,Mz]: [2.47201271e-14 1.34293272e-14 3.34105227e-14
2.02034013e-15
2.82483165e-15 2.18812796e-15]

Best local correction for the realized residual

log_mass_scale	-1
log_second_moment_cholesky_xx_scale	-0.5
log_second_moment_cholesky_yy_scale	-0.5
log_second_moment_cholesky_zz_scale	+8.381081488e-16
normalized_second_moment_cholesky_yx_offset	-0.04015201013
normalized_second_moment_cholesky_zx_offset	-0.00104405756
normalized_second_moment_cholesky_zy_offset	-0.2554596072
cog_position_body_x_m	-5.843406112e-18
cog_position_body_y_m	-1.682736653e-16
cog_position_body_z_m	-6.846111618e-17
log_force_effectiveness_1	-1
log_force_effectiveness_2	-1
log_force_effectiveness_3	-1
log_force_effectiveness_4	-1

Mean-induced parameter bias under current iid residual model

log_mass_scale	bias=-0.002070169523	std=0.02262067533
log_second_moment_cholesky_xx_scale	bias=+0.000344883378	std=0.0210755465
log_second_moment_cholesky_yy_scale	bias=+0.0002889007478	std=0.02130985288
log_second_moment_cholesky_zz_scale	bias=+1.039617847e-16	
std=6.098198033e-17		
normalized_second_moment_cholesky_yx_offset	bias=+7.783052195e-05	std=0.01185933433
normalized_second_moment_cholesky_zx_offset	bias=+0.0005419791603	std=0.04925974467
normalized_second_moment_cholesky_zy_offset	bias=-0.00225857673	std=0.03273422014
cog_position_body_x_m	bias=+0.0001192948675	std=0.001294443647
cog_position_body_y_m	bias=+0.000306441384	std=0.001575284039
cog_position_body_z_m	bias=+2.213707035e-05	std=0.001503914143
log_force_effectiveness_1	bias=-0.07750945957	std=0.05107237132
log_force_effectiveness_2	bias=-0.1319429818	std=0.05771964707
log_force_effectiveness_3	bias=-0.1846756999	std=0.06156353652
log_force_effectiveness_4	bias=-0.1311486324	std=0.05932694453

Physical parameter bias/std

mass_kg	bias=-0.00349091808	std=0.03814514881
inertia_xx_kg_m2	bias=-8.174125295e-06	std=0.0002798026188
inertia_yy_kg_m2	bias=-9.704121437e-06	std=0.000240053574
inertia_zz_kg_m2	bias=+7.791574159e-06	std=0.0003121684906
inertia_xy_kg_m2	bias=-1.805518692e-06	std=0.0002007907037
inertia_xz_kg_m2	bias=-2.727445699e-06	std=0.0002476294654
inertia_yz_kg_m2	bias=+1.203656594e-05	std=0.0001743387802

cog_position_body_x_m	bias=+0.0001192948675	std=0.001294443647
cog_position_body_y_m	bias=+0.000306441384	std=0.001575284039
cog_position_body_z_m	bias=+2.213707035e-05	std=0.001503914143
force_effectiveness_1	bias=-0.06737740745	std=0.04439618069
force_effectiveness_2	bias=-0.09431233666	std=0.04125778204
force_effectiveness_3	bias=-0.1073986129	std=0.03580242787
force_effectiveness_4	bias=-0.09068404726	std=0.04102221536

Parameter covariance in raw 14-D coordinate

```
[ 0.00051169495, 6.4820821e-05, 1.0926284e-05, -3.9273139e-19, -1.9164644e-05,
 3.4377024e-05, 7.6351024e-05, -1.3227054e-06, -1.9655498e-06, 2.7806268e-06, 0.000479633,
 0.00040782565, 0.00041427254, 0.00040705107]
[ 6.4820821e-05, 0.00044417866, -1.0148768e-05, -7.7563099e-20, -3.6028393e-05,
 0.00030284549, -0.0001317773, 3.4510021e-06, 5.888504e-07, 1.739825e-06, 3.6556084e-05,
 7.1746752e-05, 6.2294905e-05, 4.5122862e-05]
[ 1.0926284e-05, -1.0148768e-05, 0.00045410983, 1.4976169e-19, -9.3516663e-05,
 0.00030241895, -0.00027542964, -1.2885332e-06, 3.1031956e-06, -6.3352434e-06,
 -7.125808e-06, 7.8714155e-06, -4.0645556e-06, 4.4300358e-05]
[-3.9273139e-19, -7.7563099e-20, 1.4976169e-19, 3.7188019e-33, -3.0307237e-20,
 1.511036e-19, -1.0063125e-19, 8.8809145e-22, 1.0646842e-21, -5.5566171e-21, 1.612853e-18,
 -3.3706875e-18, 2.3952768e-18, -3.4851694e-18]
[-1.9164644e-05, -3.6028393e-05, -9.3516663e-05, -3.0307237e-20, 0.00014064381,
 -0.00023833776, 0.00013101976, 4.5086549e-07, -2.3384211e-07, 3.6511943e-06,
 3.7668833e-07, -3.658104e-05, 1.6175667e-05, -4.4521684e-05]
[ 3.4377024e-05, 0.00030284549, 0.00030241895, 0.00030241895, 1.511036e-19, -0.00023833776, 0.0024265224,
 -0.00063864441, -6.4859295e-06, 6.63409e-06, -2.7556352e-06, -1.1387373e-05,
 2.6532272e-05, -4.4310675e-05, 0.00014158116]
[ 7.6351024e-05, -0.0001317773, -0.00027542964, -1.0063125e-19, 0.00013101976,
 -0.00063864441, 0.0010715292, -4.7293714e-06, -1.7690581e-06, 4.7939795e-06,
 0.00013746355, -1.6364484e-05, 9.3310734e-05, 2.0454223e-05]
[-1.3227054e-06, 3.4510021e-06, -1.2885332e-06, 8.8809145e-22, 4.5086549e-07,
 -6.4859295e-06, -4.7293714e-06, 1.6755844e-06, 1.9183405e-07, 1.9083396e-07,
 -8.4369699e-06, 6.2650817e-06, 8.0449526e-06, -7.0117639e-06]
[-1.9655498e-06, 5.888504e-07, 3.1031956e-06, 1.0646842e-21, -2.3384211e-07, 6.63409e-06,
 -1.7690581e-06, 1.9183405e-07, 2.4815198e-06, 6.7833507e-07, -1.1653833e-05,
 -7.0708555e-06, 8.3826489e-06, 1.0285924e-05]
[ 2.7806268e-06, 1.739825e-06, -6.3352434e-06, -5.5566171e-21, 3.6511943e-06,
 -2.7556352e-06, 4.7939795e-06, 1.9083396e-07, 6.7833507e-07, 2.2617577e-06,
 -7.1836987e-07, -7.1310184e-07, 7.1070598e-06, 3.6511108e-06]
[ 0.000479633, 3.6556084e-05, -7.125808e-06, 1.612853e-18, 3.7668833e-07, -1.1387373e-05,
 0.00013746355, -8.4369699e-06, -1.1653833e-05, -7.1836987e-07, 0.0026083871, -0.0013859135,
 0.0027413984, -0.0014164523]
[ 0.00040782565, 7.1746752e-05, 7.8714155e-06, -3.3706875e-18, -3.658104e-05,
 2.6532272e-05, -1.6364484e-05, 6.2650817e-06, -7.0708555e-06, -7.1310184e-07,
 -0.0013859135, 0.0033315577, -0.0020351487, 0.0030975268]
[ 0.00041427254, 6.2294905e-05, -4.0645556e-06, 2.3952768e-18, 1.6175667e-05,
 -4.4310675e-05, 9.3310734e-05, 8.0449526e-06, 8.3826489e-06, 7.1070598e-06,
 0.0027413984, -0.0020351487, 0.003790069, -0.0020889056]
[ 0.00040705107, 4.5122862e-05, 4.4300358e-05, -3.4851694e-18, -4.4521684e-05,
 0.00014158116, 2.0454223e-05, -7.0117639e-06, 1.0285924e-05, 3.6511108e-06,
 -0.0014164523, 0.0030975268, -0.0020889056, 0.0035196863]
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Parameter second moment in raw 14-D coordinate

```
[ 0.00051598055, 6.4106854e-05, 1.032821e-05, -6.0794991e-19, -1.9325766e-05,
 3.3255035e-05, 8.102666e-05, -1.569666e-06, -2.5999354e-06, 2.7347993e-06, 0.00064009072,
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    0.00068096999, 0.00079658254, 0.00067855097]
[ 6.4106854e-05, 0.0004442976, -1.0049131e-05, -4.1708407e-20, -3.6001551e-05, 0.0003030324,
-0.00013255625, 3.492145e-06, 6.9453694e-07, 1.7474597e-06, 9.8243593e-06,
2.624181e-05, -1.3966745e-06, -1.0812135e-07]
[ 1.032821e-05, -1.0049131e-05, 0.00045419329, 1.7979633e-19, -9.3494178e-05,
0.00030257553, -0.00027608214, -1.2540688e-06, 3.1917268e-06, -6.328848e-06,
-2.9518349e-05, -3.0247011e-05, -5.7417503e-05, 6.4114204e-06]
[-6.0794991e-19, -4.1708407e-20, 1.7979633e-19, 1.4526855e-32, -2.2215837e-20,
2.0744872e-19, -3.3543692e-19, 1.3290199e-20, 3.2922877e-20, -3.2552078e-21,
-6.4451688e-18, -1.7087715e-17, -1.6803939e-17, -1.7119615e-17]
[-1.9325766e-05, -3.6001551e-05, -9.3494178e-05, -2.2215837e-20, 0.00014064987,
-0.00023829558, 0.00013084398, 4.6015028e-07, -2.0999162e-07, 3.6529172e-06,
-5.6559134e-06, -4.6850231e-05, 1.8022611e-06, -5.4729051e-05]
[ 3.3255035e-05, 0.0003030324, 0.00030257553, 2.0744872e-19, -0.00023829558, 0.0024268162,
-0.00063986851, -6.4212742e-06, 6.8001749e-06, -2.7436373e-06, -5.3395885e-05,
-4.4978074e-05, -0.00014440106, 7.0501335e-05]
[ 8.102666e-05, -0.00013255625, -0.00027608214, -3.3543692e-19, 0.00013084398,
-0.00063986851, 0.0010766303, -4.998808e-06, -2.4611795e-06, 4.7439812e-06,
0.00031252461, 0.00028163886, 0.00051041497, 0.00031666347]
[-1.569666e-06, 3.492145e-06, -1.2540688e-06, 1.3290199e-20, 4.6015028e-07, -6.4212742e-06,
-4.998808e-06, 1.6898156e-06, 2.2839093e-07, 1.934748e-07, -1.7683451e-05,
-9.4750388e-06, -1.3985911e-05, -2.2657123e-05]
[-2.5999354e-06, 6.9453694e-07, 3.1917268e-06, 3.2922877e-20, -2.0999162e-07,
6.8001749e-06, -2.4611795e-06, 2.2839093e-07, 2.5754261e-06, 6.8511879e-07,
-3.5405939e-05, -4.7503645e-05, -4.8209628e-05, -2.9903445e-05]
[ 2.7347993e-06, 1.7474597e-06, -6.328848e-06, -3.2552078e-21, 3.6529172e-06,
-2.7436373e-06, 4.7439812e-06, 1.934748e-07, 6.8511879e-07, 2.2622478e-06,
-2.4342022e-06, -3.6339329e-06, 3.0188809e-06, 7.4786426e-07]
[ 0.00064009072, 9.8243593e-06, -2.9518349e-05, -6.4451688e-18, -5.6559134e-06,
-5.3395885e-05, 0.00031252461, -1.7683451e-05, -3.5405939e-05, -2.4342022e-06,
0.0086161034, 0.0088409158, 0.017055512, 0.0087488073]
[ 0.00068096999, 2.624181e-05, -3.0247011e-05, -1.7087715e-17, -4.6850231e-05,
-4.4978074e-05, 0.00028163886, -9.4750388e-06, -4.7503645e-05, -3.6339329e-06,
0.0088409158, 0.020740508, 0.022331514, 0.020401668]
[ 0.00079658254, -1.3966745e-06, -5.7417503e-05, -1.6803939e-17, 1.8022611e-06,
-0.00014440106, 0.00051041497, -1.3985911e-05, -4.8209628e-05, 3.0188809e-06,
0.017055512, 0.022331514, 0.037895183, 0.02213106]
[ 0.00067855097, -1.0812135e-07, 6.4114204e-06, -1.7119615e-17, -5.4729051e-05,
7.0501335e-05, 0.00031666347, -2.2657123e-05, -2.9903445e-05, 7.4786426e-07,
0.0087488073, 0.020401668, 0.02213106, 0.02071965]

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Physical parameter covariance

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[ 0.0014550524, 6.4478887e-07, 1.7259905e-06, 9.0333813e-07, 4.0275304e-07,
-2.9248625e-07, -9.2887935e-07, -2.2304725e-06, -3.3144983e-06, 4.6889592e-06,
0.0007030762, 0.00049157579, 0.00040626444, 0.00047462408]
[ 6.4478887e-07, 7.8289505e-08, 9.9988887e-09, 4.5574945e-08, -1.4516258e-08,
1.0116453e-08, -3.0900933e-08, -3.9574442e-08, 2.9072384e-08, -1.9499224e-08,
6.0023764e-07, -2.6188389e-07, 3.7398385e-07, 1.7253518e-07]
[ 1.7259905e-06, 9.9988887e-09, 5.7625718e-08, 1.2098327e-08, -1.1580765e-08,
4.1229688e-09, -2.3378348e-08, 4.2746694e-09, -4.5633172e-09, 4.2999464e-08,
9.679617e-07, 3.9945257e-07, 6.3694734e-07, 3.6596053e-07]
[ 9.0333813e-07, 4.5574945e-08, 1.2098327e-08, 9.7449167e-08, -2.8239551e-09,
-2.1724236e-08, -7.0805256e-10, 1.8806036e-08, 4.4326657e-08, -3.0911054e-08,
2.0921612e-07, 2.6976282e-07, 3.9477458e-07, 3.7304255e-07]
[ 4.0275304e-07, -1.4516258e-08, -1.1580765e-08, -2.8239551e-09, 4.0316907e-08,

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-1.8403104e-08, 2.0547689e-09, -1.2455772e-08, 3.1919006e-09, -6.4861311e-08,
-4.9264051e-08, 3.7692067e-07, -2.1072704e-07, 4.838525e-07]
[-2.9248625e-07, 1.0116453e-08, 4.1229688e-09, -2.1724236e-08, -1.8403104e-08,
6.1320352e-08, 3.674073e-09, 3.2560141e-08, -3.3347147e-08, 1.3830746e-08,
4.9415017e-08, -9.5851488e-08, 1.2912708e-07, -4.9233234e-07]
[-9.2887935e-07, -3.0900933e-08, -2.3378348e-08, -7.0805256e-10, 2.0547689e-09,
3.674073e-09, 3.039401e-08, 4.2681911e-08, -8.9647205e-09, -5.4415953e-09, -6.9386073e-07,
2.5946603e-08, -2.8449854e-07, -3.2927917e-07]
[-2.2304725e-06, -3.9574442e-08, 4.2746694e-09, 1.8806036e-08, -1.2455772e-08,
3.2560141e-08, 4.2681911e-08, 1.6755844e-06, 1.9183405e-07, 1.9083396e-07,
-7.3340875e-06, 4.4782563e-06, 4.6785622e-06, -4.848355e-06]
[-3.3144983e-06, 2.9072384e-08, -4.5633172e-09, 4.4326657e-08, 3.1919006e-09,
-3.3347147e-08, -8.9647205e-09, 1.9183405e-07, 2.4815198e-06, 6.7833507e-07,
-1.0130442e-05, -5.0542204e-06, 4.8749503e-06, 7.112306e-06]
[4.6889592e-06, -1.9499224e-08, 4.2999464e-08, -3.0911054e-08, -6.4861311e-08,
1.3830746e-08, -5.4415953e-09, 1.9083396e-07, 6.7833507e-07, 2.2617577e-06,
-6.2446442e-07, -5.0972245e-07, 4.1331283e-06, 2.5245974e-06]
[0.0007030762, 6.0023764e-07, 9.679617e-07, 2.0921612e-07, -4.9264051e-08, 4.9415017e-08,
-6.9386073e-07, -7.3340875e-06, -1.0130442e-05, -6.2446442e-07, 0.0019710209,
-0.00086114823, 0.0013858642, -0.00085139027]
[0.00049157579, -2.6188389e-07, 3.9945257e-07, 2.6976282e-07, 3.7692067e-07,
-9.5851488e-08, 2.5946603e-08, 4.4782563e-06, -5.0542204e-06, -5.0972245e-07,
-0.00086114823, 0.0017022046, -0.00084599399, 0.001530962]
[0.00040626444, 3.7398385e-07, 6.3694734e-07, 3.9477458e-07, -2.1072704e-07,
1.2912708e-07, -2.8449854e-07, 4.6785622e-06, 4.8749503e-06, 4.1331283e-06,
0.0013858642, -0.00084599399, 0.0012818138, -0.00083999144]
[0.00047462408, 1.7253518e-07, 3.6596053e-07, 3.7304255e-07, 4.838525e-07,
-4.9233234e-07, -3.2927917e-07, -4.848355e-06, 7.112306e-06, 2.5245974e-06,
-0.00085139027, 0.001530962, -0.00083999144, 0.0016828222]

Physical parameter second moment

[0.0014672389, 6.7332407e-07, 1.7598668e-06, 8.7613838e-07, 4.0905595e-07,
-2.8296496e-07, -9.7089802e-07, -2.6469212e-06, -4.3842601e-06, 4.6116805e-06,
0.00093828521, 0.00082081244, 0.0007811842, 0.00079119466]
[6.7332407e-07, 7.8356322e-08, 1.0078211e-08, 4.5511256e-08, -1.45015e-08, 1.0138748e-08,
-3.0999321e-08, -4.0549574e-08, 2.6567494e-08, -1.9680175e-08, 1.150989e-06,
5.0903697e-07, 1.2518736e-06, 9.1379795e-07]
[1.7598668e-06, 1.0078211e-08, 5.7719888e-08, 1.2022716e-08, -1.1563244e-08,
4.1494363e-09, -2.3495152e-08, 3.1170175e-09, -7.5370616e-09, 4.2784643e-08,
1.6218002e-06, 1.3146709e-06, 1.6791565e-06, 1.2459695e-06]
[8.7613838e-07, 4.5511256e-08, 1.2022716e-08, 9.7509875e-08, -2.838023e-09,
-2.1745487e-08, -6.1426877e-10, 1.9735531e-08, 4.6714317e-08, -3.0738571e-08,
-3.1575995e-07, -4.6507875e-07, -4.4202968e-07, -3.3352893e-07]
[4.0905595e-07, -1.45015e-08, -1.1563244e-08, -2.838023e-09, 4.0320167e-08, -1.839818e-08,
2.0330367e-09, -1.2671161e-08, 2.6386149e-09, -6.490128e-08, 7.2387118e-08,
5.4720336e-07, -1.6816837e-08, 6.4758424e-07]
[-2.8296496e-07, 1.0138748e-08, 4.1494363e-09, -2.1745487e-08, -1.839818e-08,
6.1327791e-08, 3.641244e-09, 3.2234771e-08, -3.4182949e-08, 1.3770369e-08,
2.3318324e-07, 1.6138029e-07, 4.2205096e-07, -2.4499652e-07]
[-9.7089802e-07, -3.0999321e-08, -2.3495152e-08, -6.1426877e-10, 2.0330367e-09,
3.641244e-09, 3.0538889e-08, 4.4117811e-08, -5.2762185e-09, -5.175141e-09, -1.5048533e-06,
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